Form C

Core Focus on Proportions & Probability

Block 4 Review ~ Probability and Random Sampling

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. Of the first 36 customers at a smoothie stand, 9 ordered a strawberry smoothie. What is the experimental probability that the next customer will order a strawberry smoothie?

A. $\frac{1}{9}$ B. $\frac{1}{4}$ C. $\frac{1}{3}$ D. $\frac{3}{4}$

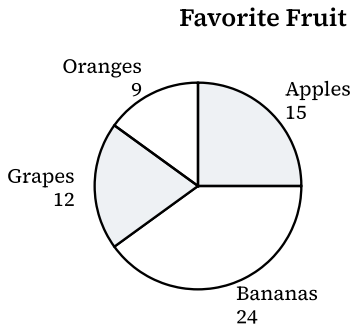
B ($\frac{9}{36} = \frac{1}{4}$)

2. A bag holds 6 red, 3 white and 1 blue marble. A marble is drawn and put back 50 times. For 2a – 2c, circle YES or NO to show whether you would expect the given color to be drawn the number of times stated.

2a. 30 times red	YES	NO
2b. 10 times white	YES	NO
2c. 5 times blue	YES	NO

2a YES ($\frac{6}{10} \cdot 50 = 30$) · 2b NO ($\frac{3}{10} \cdot 50 = 15$) · 2c YES ($\frac{1}{10} \cdot 50 = 5$)

3. Omar asked 60 people to name their favorite fruit. The results are shown in the pie chart. If Omar asked 500 people, how many would you predict would choose bananas?



A. 24 people

B. 120 people

C. 200 people

D. 12,000 people

C ($\frac{24}{60} = \frac{2}{5}$; $\frac{2}{5} \cdot 500 = 200$)

4. Ezra flips a coin and then rolls a number cube. He makes a list of all possible outcomes. How many outcomes on his list show tails and a number greater than 2?

A. 2

B. 4

C. 6

D. 12

B (T3, T4, T5, T6 out of 12 outcomes)

5. A spinner has four equal sections numbered 1, 2, 3 and 4. You spin it two times. Which of the theoretical probabilities are accurate? Circle all that apply.

A. P(3 then 3) = $\frac{1}{16}$

B. P(odd number then even number) = $\frac{1}{4}$

C. P(1 then 5) = $\frac{1}{20}$

D. P(number less than 4 then 2) = $\frac{3}{16}$

E. P(even number then 4) = $\frac{1}{4}$

A, B, D. (A: ¼ · ¼ = $\frac{1}{16}$; B: ½ · ½ = ¼; C is 0, there is no 5; D: ¾ · ¼ = $\frac{3}{16}$; E: ½ · ¼ = ⅛, not ¼)

6. Nine dogs are in a pet show. Ribbons are given for first place and second place. In how many different ways can the two ribbons be awarded?

A. 17 B. 18 C. 72 D. 81

C (9 choices for first · 8 left for second = 72)

7. A suitcase lock has three wheels. Each wheel shows one of the letters A – F. What is the probability the lock's code is F-F-F?

A. $\frac{1}{6}$

B. $\frac{1}{18}$

C. $\frac{1}{36}$

D. $\frac{1}{216}$

D (6 · 6 · 6 = 216 codes, only one is F-F-F)

8. The mayor wants to know whether the people in her city support building a new skate park. Which sample will give her the best chance of an accurate prediction?

A survey of 200 residents _____.

A. at the skateboard shop downtown.

B. who call in to a radio talk show.

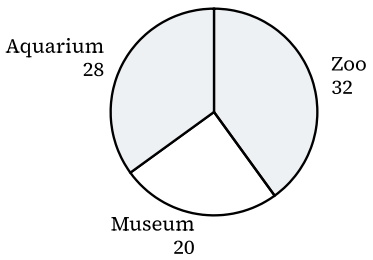
C. who live on the mayor's street.

D. picked at random from a list of all the city's residents.

D (a random sample of the whole city; the others are biased or only include volunteers)

9. Eighty students voted on where to go for a field trip. The results are below. Which inference is the weakest?

Field Trip Vote



A. The zoo is the most popular choice.

B. More students chose the aquarium than the museum.

C. If the choices were 'Zoo' or 'Museum or Aquarium', most students would choose the zoo.

D. Fewer than half of the students chose the zoo.

C (Museum or Aquarium has 20 + 28 = 48 votes, more than the zoo's 32)

10. The five-number summary and mean for two data sets are shown below. Each data set has 9 values. Which statement is true?

Set X: 12 ~ 14 ~ 15 ~ 16 ~ 18
mean = 15

Set Y: 2 ~ 8 ~ 15 ~ 22 ~ 28
mean = 15

A. Set X has the greater mean absolute deviation.

B. Set Y has the greater mean absolute deviation.

C. The sets have equal mean absolute deviations because their means are equal.

D. Cannot be determined

B (Set Y's values are spread much farther from the mean of 15)

11. The minutes Priya and Sam practiced piano on eight days are shown below. Which statement best compares their median times?

Priya: 30, 25, 40, 35, 20, 45, 30, 35
Sam: 40, 30, 35, 25, 45, 40, 30, 50

A. Their medians are equal.

B. Priya's median is 5 minutes greater than Sam's.

C. Sam's median is 5 minutes greater than Priya's.

D. Sam's median is 2.5 minutes greater than Priya's.

C (in order, Priya: 20, 25, 30, 30, 35, 35, 40, 45 → 32.5; Sam: 25, 30, 30, 35, 40, 40, 45, 50 → 37.5)

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Part II – Constructed Response

A spinner has 12 equal sections numbered 1 to 12. Leah spins it once. Find each probability for the number she spins.

1. P(9)

$\frac{1}{12}$

2. P(number greater than 8)

$\frac{4}{12} = \frac{1}{3}$ (9, 10, 11, 12)

3. P(number less than 13)

1 (certain, every section is less than 13)

4. P(not a multiple of 4)

$\frac{9}{12} = \frac{3}{4}$ (4, 8, 12 are multiples of 4)

5. A box has 12 chocolate, 10 peanut butter and 8 oatmeal granola bars. Luis takes one bar without looking. Which flavor has a $\frac{1}{3}$ chance of being chosen? Explain how you know your answer is correct.

Peanut butter. There are 12 + 10 + 8 = 30 bars and P(peanut butter) = $\frac{10}{30} = \frac{1}{3}$. (Chocolate is $\frac{12}{30} = \frac{2}{5}$ and oatmeal is $\frac{8}{30} = \frac{4}{15}$.)

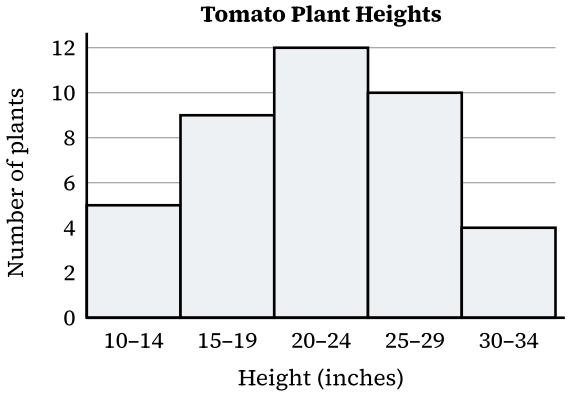
6. A park ranger wanted to estimate the number of fish in a pond. She caught 20 fish, marked them and put them back. Two weeks later she caught 25 fish, and 5 of them were marked. Estimate the number of fish in the pond. Show all work necessary to justify your answer.

100 fish. $\frac{5 \text{ marked}}{25 \text{ caught}} = \frac{20 \text{ marked}}{n \text{ total}}$; $5n = 500$, $n = 100$. (Check: 5 of 25 is $\frac{1}{5}$, and $\frac{20}{100} = \frac{1}{5}$.)

7. A spinner has 8 equal sections, and 3 of them are green. Kenji spins it 64 times. How many times would you predict it lands on green? Explain your reasoning.

24 times. P(green) = $\frac{3}{8}$, and $\frac{3}{8} \cdot 64 = 24$.

8. The histogram shows the heights of the tomato plants in a community garden.



a. How many tomato plants are in the garden?

5 + 9 + 12 + 10 + 4 = 40 plants

b. Find the probability that a randomly chosen plant is at least 25 inches tall.

$\frac{10 + 4}{40} = \frac{14}{40} = \frac{7}{20}$

9. Mia orders one cup of frozen yogurt. She chooses one flavor (vanilla, berry, mango or chocolate) and one topping (sprinkles or nuts). How many different cups are possible? Show a tree diagram, list or table to support your answer.

8 cups (4 · 2). List: vanilla–sprinkles, vanilla–nuts, berry–sprinkles, berry–nuts, mango–sprinkles, mango–nuts, chocolate–sprinkles, chocolate–nuts.

10. Grace takes a multiple choice quiz with four questions. Each question has five possible answers. If Grace guesses on every question, what is the probability she answers them all correctly?

$\frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{5} = \frac{1}{625}$

11. Andre flips a coin. He then draws a marble from a bag with 3 red marbles and 7 blue marbles. What is the probability Andre flips heads and draws a blue marble?

$\frac{1}{2} \cdot \frac{7}{10} = \frac{7}{20}$

12. Kai wants to know whether students at his school want a new gym. He asks the first 30 students who arrive at the after-school basketball practice. Since 27 of them say yes, he decides that 90% of the students at his school want a new gym. Is his prediction accurate based on the sample surveyed? Explain your reasoning.

No. The sample is biased: students at basketball practice use the gym and are more likely to want a new one. He should survey a random sample of all the students at the school.

13. Two different random samples were used to find students' favorite after-school activity. The results are below. Make at least two inferences.

	Sports	Music	Art	Games	TOTAL
Sample 1	45	15	10	20	90
Sample 2	40	20	10	30	100

Sample answers: Sports is the most popular activity (45 of 90 = 50% and 40 of 100 = 40%). Art is the least popular (about 11% and 10%). Games is the second favorite in both samples (about 22% and 30%).

Use the data below to answer numbers 14 – 20. The data shows the points scored by two basketball teams, the Hawks and the Owls, in their first 9 games.

Hawks: 47, 33, 53, 40, 57, 45, 37, 50, 43
Owls: 49, 44, 52, 46, 48, 51, 46, 47, 49

14. Find the mean of each data set.

Hawks: $\frac{405}{9}$ = 45 points. Owls: $\frac{432}{9}$ = 48 points.

15. Find the median of each data set.

Hawks: 33, 37, 40, 43, <u>45</u>, 47, 50, 53, 57 → 45.
Owls: 44, 46, 46, 47, <u>48</u>, 49, 49, 51, 52 → 48.

16. Ruby said the Owls' median score was 4 points higher than the Hawks' median score. Is her comparison accurate? Explain your reasoning.

No. The medians are 48 and 45, so the Owls' median is only 48 − 45 = 3 points higher.

17. Find the five-number summary of each data set.

Hawks: 33 ~ 38.5 ~ 45 ~ 51.5 ~ 57. Owls: 44 ~ 46 ~ 48 ~ 50 ~ 52. (Leave out the median: Q1 is the median of the lowest 4 values, Q3 of the highest 4.)

18. Find the interquartile range of each data set.

Hawks: 51.5 − 38.5 = 13. Owls: 50 − 46 = 4.

19. Find the mean absolute deviation of each data set.

Hawks (mean 45): distances 12, 8, 5, 2, 0, 2, 5, 8, 12; sum 54; MAD = $\frac{54}{9}$ = 6 points. Owls (mean 48): distances 4, 2, 2, 1, 0, 1, 1, 3, 4; sum 18; MAD = $\frac{18}{9}$ = 2 points.

20. Write one sentence comparing a measure of variability between the two data sets.

Sample: The Hawks' MAD (6) is three times the Owls' MAD (2), so the Hawks' scores vary much more from game to game. (Or: the Hawks' IQR is 13 and the Owls' is 4.)

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Part III – Problem Solving

Ella asked a random sample of 150 students at her school to name their favorite season and their favorite pizza topping. The results are below.

Favorite Season	Number of Students
Summer	60
Fall	45
Winter	15
Spring	30

Favorite Topping	Number of Students
Cheese	50
Pepperoni	75
Veggie	25

There are 900 students at Ella's school. Based on Ella's data, what is the probability that the next student she asks will prefer fall and cheese pizza? How many students in the whole school would you predict prefer both fall and cheese pizza?

P(fall) = $\frac{45}{150} = \frac{3}{10}$. P(cheese) = $\frac{50}{150} = \frac{1}{3}$. P(fall and cheese) = $\frac{3}{10} \cdot \frac{1}{3} = \frac{3}{30} = \frac{1}{10}$. Prediction: $\frac{1}{10} \cdot 900 = 90$ students.